ECO 202 — In-Class Exam 2

Worked solutions

Fall 2026 | Classes 6–9

Equivalent exact expressions and valid alternative reasoning are acceptable. For practice study, attempt the paper first, then compare reasoning, close these solutions, and reconstruct any missed method independently.

Multiple-choice key

Question	Answer	Reason and distractor audit
1	A	Response related to the outcome can change who is observed. B assigns a direction without a mechanism; C confuses selected and responding samples; D wrongly treats sample size as a cure for selection.
2	A	The randomized object is the offer, and the direct study scope is the participants. B confuses offer with receipt and average with individual effects; C assumes external validity; D promises exact realized balance.
3	D	The proposed union is $3/4 + 3/4 - 1/4 = 5/4 > 1$. A describes independent fair events, B disjoint fair events, and C identical fair events; each is possible.
4	D	Bayes' rule uses the base rate and both conditional flag rates. A reverses the conditional; B is the already-known complement 1/5 and adds no missing base-rate information; C supplies scale but not event composition.

5. Equal inclusion and simple random sampling

(a) The chance mechanism and probabilities of all possible samples are specified, so it is a probability sample. A appears in ABC, DEA, EAB, giving inclusion probability 3/5.

(b) B appears in ABC, BCD, EAB; C in ABC, BCD, CDE; D in BCD, CDE, DEA; E in CDE, DEA, EAB. Together with (a), this gives three appearances and selection probability 3/5 for each firm.

It is not an SRS. For example, ABD has probability zero here, while an SRS would give all ten triples probability 1/10. Equal individual inclusion probabilities do not make the full sets of possible samples the same.

(c) The population mean of the five firms is

$$\mu = \frac{y_A + y_B + y_C + y_D + y_E}{5}.$$

The observed statistic is $\bar{y} = (y_A + y_B + y_C)/3$. No: inclusion probabilities describe the chance mechanism across repetitions, not an equality between the values in one sample and the population. Since $\mu = (3\bar{y} + y_D + y_E)/5$, the two means are equal only if $(y_D + y_E)/2 = \bar{y}$, that is, only if the two omitted firms happen to average the same employment as the three included firms.

(d) The sample mean in (c) uses only council members, so nonmembers are missing from the sampling frame: undercoverage of the citywide target. No: observing all five members removes sampling variability within the frame but supplies no information on omitted nonmembers.

6. An experiment and its causal target

(a) The effects are $2, 1, 3, 2$ correctly solved exercises for A–D, respectively. Their average is $(2 + 1 + 3 + 2)/4 = 2$ exercises per volunteer.

(b) When A and D receive the offer, observed outcomes in A–D order are $5, 5, 2, 10$. The offered mean is $(5 + 10)/2 = 15/2$ and the unoffered mean is $(5 + 2)/2 = 7/2$. The difference is 4 exercises.

(c) *(i)* When B and C receive the offer, observed outcomes in A–D order are $3, 6, 5, 8$. The offered mean and the unoffered mean both equal $11/2$, so the offered-minus-unoffered difference is zero.

(ii) The difference is zero while the fixed ATE is 2. The offered pair's mean no-offer outcome is $(5 + 2)/2 = 7/2$, compared with $(3 + 8)/2 = 11/2$ for the unoffered pair. Neither fact proves that randomization failed. Different people supply the two observed means, and random assignment can produce chance imbalance in one realization. Failure would concern the assignment procedure or its execution, not merely an imbalance in the assignment that occurred.

(iii) Averaging the offered-minus-unoffered difference over all six assignments recovers the ATE, 2 exercises per volunteer.

(d) Comparison (1), comparing all volunteers by assigned offer, retains the randomized assignment and estimates the average effect of the offer for these volunteers under its implementation (intention to treat). Attendance is chosen after assignment and can depend on motivation or other outcome-related characteristics. Comparison (2), comparing attendees and nonattendees, is not automatically randomized and does not by itself identify the causal effect of attending tutoring.

7. Events, overlap, and independence

(a) At least one: $3/5 + 1/2 - 1/5 = 9/10$. Neither: $1 - 9/10 = 1/10$. Exactly one is the disjoint union $(A \cap B^c) \cup (A^c \cap B)$, with probability

$$\left(\frac{3}{5} - \frac{1}{5}\right) + \left(\frac{1}{2} - \frac{1}{5}\right) = \frac{7}{10}.$$

(b) $\mathbb{P}(B \mid A) = (1/5)/(3/5) = 1/3$: the conditioning event is A, so the denominator is $\mathbb{P}(A) = 3/5$. Conversely, $\mathbb{P}(A \mid B) = (1/5)/(1/2) = 2/5$: the conditioning event is B, with denominator $\mathbb{P}(B) = 1/2$. The intersection is the same, but the conditioning events, and hence the denominators, differ.

(c) They are not disjoint because $\mathbb{P}(A \cap B) = 1/5 > 0$. They are not independent because $\mathbb{P}(A)\mathbb{P}(B) = 3/10 \neq 1/5$. Equivalently, $\mathbb{P}(B \mid A) = 1/3 \neq 1/2 = \mathbb{P}(B)$.

8. Updating the likely source of a shipment

(a) The South–late probability is $(3/5)(7/10) = 21/50$. Subtraction within each source gives

	Late L	Not late L^c	Total
North N	5/50	15/50	20/50
South N^c	21/50	9/50	30/50
Total	26/50	24/50	1

Every cell is nonnegative, and the row totals and the column totals each sum to one.

(b) Among late shipments, the North fraction is

$$\mathbb{P}(N \mid L) = \frac{5/50}{26/50} = \frac{5}{26}.$$

The late fraction among North shipments is instead $\mathbb{P}(L \mid N) = (1/10)/(2/5) = 1/4$. One conditions on lateness; the other conditions on source.

(c) They are not independent: $\mathbb{P}(L \mid N) = 1/4 \neq 13/25 = \mathbb{P}(L)$. Learning lateness lowers the probability of North, since $5/26 < 2/5$ (cross-multiplication gives $25 < 52$). South's late rate is higher, so a late shipment provides evidence in favor of South.